

Mobile Phone Teardown Photo Capture Protocol

A step-by-step guide for capturing training images of end-of-life mobile phones for the battery segmentation dataset (YOLOv11n-seg / CVAT.ai pipeline).

Document owner	Computer Vision Team, Redback Operations
Applies to	Anyone capturing new phone images for the battery segmentation dataset (Project 7)
Status	Living document — update as the dataset and model requirements evolve
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1. Purpose & Scope

This document sets out a repeatable process for photographing end-of-life mobile phones so the images can be used to expand and diversify the battery segmentation training dataset. It exists so that:

- Anyone on the Computer Vision team (this trimester or in future trimesters) can capture new phone images without re-deriving the method from scratch.
- Images contributed to the dataset are consistent enough in coverage (lighting, background, angle) to be useful for training and evaluation.
- The reasoning behind the capture matrix (why 20 images, why these variations) is preserved as project knowledge, not just as a one-off instruction from a mentor.

This protocol covers photo capture only.

2. Ethics & Consent

Before photographing a donated or sourced device, confirm the following:

1. The donor is aware the device (or images of it) will be used for a computer vision training dataset associated with Redback Operations / Deakin University.
2. Written or recorded consent has been obtained from the donor for use of teardown images in this context.
3. Any personal data on or in the device (SIM cards, SD cards, stored user data) has been removed or the donor has been informed it will not be recovered or accessed.
4. The device and any identifying marks (e.g. custom engravings, stickers with personal information) are checked before photographing and excluded or obscured if needed.

Keep a simple record of consent (who, when, how obtained) alongside the dataset .

3. Equipment & Setup

3.1 Equipment checklist

- Camera or smartphone capable of at least 12MP resolution, with autofocus
- At least two distinct background surfaces (see Section 4)
- Access to at least two different lighting setups (see Section 4)

3.2 Camera settings

- Use the highest resolution setting available on the device
- Keep resolution and aspect ratio consistent across the full set of 20 images for a given phone
- Disable heavy auto-filters or beautification modes — the dataset needs realistic, unprocessed images
- Avoid digital zoom; move the camera physically closer instead

4. Capture Matrix — 20 Images per Phone

The rule of thumb is 20 photos per phone, varying lighting, background, rotation, and angle. The matrix below is a recommended way to structure those 20 shots so coverage is systematic rather than ad hoc.

Variation type	Recommended options	Purpose
Lighting	2–3 conditions, e.g. bright/even indoor light, dim/warm light, natural daylight near a window	Prevents the model from relying on lighting cues that won't hold in a real recycling environment
Background	2 surfaces, e.g. plain neutral background and a cluttered/textured surface (bench, workbench, other e-waste)	Improves generalisation so the model isn't relying on background as a shortcut
Rotation	Phone rotated in-plane at roughly 0°, 90°, 180°, 270°	Ensures the model doesn't assume a fixed orientation for the device or battery

Camera angle	Straight-on (top-down), and 2–3 oblique angles (~30–45°) from different sides	Captures how the device and exposed battery look from angles a robotic arm's camera might see
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Example 20-shot breakdown (adapt as needed):

- 2 lighting conditions × 2 backgrounds × 5 angle/rotation combinations = 20 images

5. Step-by-Step Capture Procedure

1. Confirm consent and remove/ignore personal data as per Section 2.
2. Remove the phone's back cover safely.
3. Take the first reference photo.
4. Rotate the phone through the planned rotation steps (e.g. 0°, 90°, 180°, 270°), photographing at each step.
5. Adjust the camera to each planned oblique angle and repeat photography at that background/lighting combination.
6. Switch to the next background and/or lighting condition and repeat steps 4–5 until 20 images are captured.
7. Review all 20 images immediately on-device for focus, framing, and exposure issues before disassembling or returning the phone.
8. Transfer images to the shared dataset storage location and rename them to match the convention in Section 6, using the rename.py script in Appendix B.
9. Begin annotating the images in CVAT.ai.

6. File Naming & Metadata Logging

6.1 File naming convention

Use a consistent, sortable naming pattern so images can be traced back to a device and capture session without opening each file:

DeviceModelName_Indoor/Outdoor_Lighting_SeqNo.jpg

Example: Microsoft_Lumia_640_indoor_dim_2.jpg

6.1.1 Lighting tags used in this dataset

In practice, the <Lighting> field settled on four values across the dataset collected so far. Use one of these exact tags (lowercase, underscore-separated) so filenames stay machine-sortable and consistent with the existing 184-image set:

Tag	Description
indoor_cool	Indoor, bright/even (cool-toned, e.g. LED or fluorescent) lighting.
indoor_warm	Indoor, warm-toned (e.g. incandescent/tungsten) lighting.
indoor_dim	Indoor, low-light / dim conditions.
outdoor_daylight	Outdoor, natural daylight.

7. Quality Checklist — Before Submitting a Photo Set

- All 20 images present and correctly named
- At least two lighting conditions represented
- At least two backgrounds represented

- Full rotation range covered (not just front-facing shots)
- Straight-on and oblique angles both included
- No personal data visible (screens, SIM/SD cards, engraved names)

Appendix A — Sample Images by Lighting Condition

One representative example is kept here for each of the four lighting tags defined in Section 6.1.1, so a new team member can see what each tag looks like in practice before shooting their own set. Insert one clear, correctly-named example image per box below.

indoor_cool

Example filename: Microsoft_Lumia_640_indoor_cool_01.jpeg



indoor_warm

Example filename: Microsoft_Lumia_640_indoor_warm_10.jpeg



indoor_dim

Example filename: *Samsung_Galaxy_S5_indoor_dim_04.jpeg*



outdoor_daylight

Example filename: *Samsung_Omnia_7_outdoor_daylight_13.jpeg*



Appendix B — Dataset Batch Renaming Script

Once a photo set for a phone has been captured and reviewed, the file names in Section 6.1 are applied in bulk rather than typed one by one. The Python script below (rename.py) sorts every image in a folder and renames it "<base_name>_<seq><ext>", where the base name is typed in at the prompt (e.g. "Microsoft Lumia 640 indoor_cool") and the sequence number is zero-padded automatically. Update the "folder" path in the script to point at the OneDrive folder for the phone being processed before running it.

Script in use

```
import os

def rename_images(folder_path, base_name):
    extensions = ('.jpg', '.jpeg', '.png', '.gif', '.bmp', '.webp', '.tiff')

    # Get all image files in the folder
    images = [f for f in os.listdir(folder_path) if f.lower().endswith(extensions)]
    images.sort() # keeps a consistent order (alphabetical/by original name)

    pad_width = max(2, len(str(len(images))))

    for i, filename in enumerate(images, start=1):
        ext = os.path.splitext(filename)[1]
        number = str(i).zfill(pad_width)
        new_name = f"{base_name}_{number}{ext}"

        old_path = os.path.join(folder_path, filename)
        new_path = os.path.join(folder_path, new_name)

        os.rename(old_path, new_path)
        print(f"Renamed: {filename} -> {new_name}")

if __name__ == "__main__":
    #set below the folder path where YOUR images are located
    folder = r"C:\Documents\T2 2026\Capstone A - SIT764\Mobile Phone Tear Down Images\ZTE Boost Zume 5"

    name = input("Enter the base name for renaming: ").strip()
    rename_images(folder, name)
```

8. Version History

Version	Date	Change
v1.0	Trimester 2, 2026	Initial protocol drafted based on mentor guidance (20 images per phone, varied lighting/background/rotation/angle)
v1.1	25 Sep 2026	Documented the four finalised lighting tags (Section 6.1.1), added a sample image gallery (Appendix A) and the dataset batch-renaming script with usage notes (Appendix B). Prepared by Senara Perera for handover to the next cohort.